赵浩翰

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教育经历

南京大学 *金融学* *2021.09 - 2025.06*
GPA: 4.58/5.0
相关课程：Python 程序设计 | 数据结构与算法分析 | 机器学习 | 常微分方程 | 随机过程 | 行为金融学 | 金融工程学

实习经历

兴业证券金工组 *（股票高频研究员）* *2024.04 - 至今*

- 根据集合竞价成交明细数据，计算个股 3s 切片的成交价量，以计算并回测兴业金工高频因子；
- 日内股价跳跃预测，使用 RF、NN 等模型，结合流动性和技术指标，以 5 分钟为窗口预测日内股价跳跃；

浙江量衍投资管理有限公司 *（期货中低频研究员）* *2023.06 - 2023.09*

- 日常工作：维护公司数据库，进行期货指数编制、因子计算等数据处理、分析任务；
- 业绩归因回测研究：研究单品种多空收益、跨期套利，多品种套利、收益相关性，以聚类实现品种分类，以 rolling volatility 为品种赋权，研究不同周期、类型的策略在不同经济周期的时间效益及策略相关性；
- 动量策略市场择时：基于 MACD 和 ATR 指标构建市场波段划分体系，构建高低点识别、市场择时模型。实现全品种 sharpe 平均提升 0.04，max drawdown 平均下降 0.78，calmar 平均提升 0.01。

研究经历

VGC-GAN: A Multi-Graph Convolution Adversarial Network *（Research Assistant）* *2023.12 - 至今*

- 以 VMD-GA 算法分解股价序列，以 Pearson、Spearman、DTW 刻画股票相关性；
- 以 Multi-Graph Convolution Net、GRU 作为 Generator，同时刻画股票的时序依赖和空间依赖；
- 以 Convolution Net 作为 Discriminator，构建 WGAN 网络，进行对抗性训练，实现股价预测。

项目经历

九坤投资量化交易联赛 *（Top 30）* *2023.12*

- 使用 level-2 数据以预期收益率为目标训练 LightGBM，以即时 LOB 数据识别动量信号，参与模拟交易；
- 根据成交情况维护外挂单和持仓表，从而实现挂单处理、仓位控制，并设置止损逻辑，优化交易行为；
- 设置逻辑判断模型运行时间，根据模拟市场实际交易情况调整模型参数，日均收益 6k，决赛排名 Top 30；

衍盛私募证券投资基金管理有限公司 *（因子挖掘）* *2023.11 - 2023.12*

- 依据所给日频量价数据、基本面数据，阅读研报、论文，结合自己的理解和创新，挖掘 Alpha；
- 借鉴聚宽 factor_analyzer 库，编写因子回测框架，包括因子处理、回测指标计算、分层回测等；
- 最终个人产出 11 个因子，3 个因子质量优异（5 日 IC > 0.05，多头收益良好，分层效果较佳）。

2023 年中国大学生数学竞赛 C 题 *（省三）* *2023.09*

- 分析商超 3 年 90 万条销售数据及商品关联，完成数据处理，在天、月、年等维度实现相关性层次聚类；
- 以多项式回归量化利润和成本，以 ARIMA 预测成本，建立利润最大化非线性规划方程并以遗传算法求解；

2023 年美国大学生数学竞赛 C 题 *（M 奖）* *2023.02*

- 独立完成 359 条 12 维数据的处理、并挖掘 8 个维度的特征因子；
- 建立 ARIMA 模型和 BP 神经网络，预测游戏未来日活量，与历史逻辑走势相同；
- 依据多元回归实现单词难度评分，以 fisher 线性判别和层次聚类实现单词难度分类，准确率达 81.6%。

技能及荣誉

编程语言	Python | Pytorch | C | Git | MATLAB | R | LaTeX
技能证书	WorldQuant Alphathon Gold Certificate；WorldQuant Consultant；Coursera Machine Learing
语言成绩	IELTS 7 | CET - 6 - 612
奖学金	国家励志奖学金 × 2；恒芳奖学金 × 2

Haohan Zhao

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Education

Nanjing University *Finance* *2021.09 - 2025.06*

GPA: 4.58 / 5.0

Courses: Python | Data Structure and Algorithms | Stochastic Processes | Behavioral Finance | Financial Engineering

Intern experience

XingYe Securities Financial Engineering Group *(Stocks High Freq Quant Researcher)* *2024.04 - present*

- Backtest high-freq factors based on price and volume computed from 3s transaction records during opening auction.
- Stock price jump prediction in 5-minute windows using RF, NN, etc., combining liquidity and technical indicators. Construct a standard framework from jumps detecting to classes balancing, models training performance evaluation.

Liangyan Investment Company *(Futures Quant Researcher)* *2023.06 - 2023.09*

- Maintain database, carry out data processing and analysis tasks such as futures index compilation;
- Strategy backtest. Research the relationships, classification, and weighting among various futures varieties and strategies.
- Market timing. Construct a Market band division system based on MACD, ATR, etc. to recognize high-low points. The average sharpe increased by 0.04, max drawdown decreased by 0.78, and calmar increased by 0.01.

Research Experience

VGC-GAN: A Multi-Graph Convolution Adversarial Network *(Research Assistant)* *2023.12 - present*

- Use VMD to decompose stock price series to make them more stationary, in which GA is used to optimize its parameters. And use Pearson, Spearman and DTW to depict stock correlation.
- Multi-Graph CNN and GRU are used as Generator to simultaneously depict the temporal and spatial dependence of stocks. CNN is used as Discriminator to judge whether a sample is ture or not.
- Combine Generator and Discriminator to construct WGAN framework, add MSE Loss to original Value Function, so as to implement adversarial training and price prediction.

Projects

UBIQUANT Quant Trading Contest *(Top 30)* *2023.12*

- Use level-2 data to train LightGBM, identify trade signals with real-time LOB data during simulated trading.
- Use orders and position list to process orders, control position and stop loss, optimize trading behaviors.
- Judge model's running time, adjust running parameters according to time-changing trading situations.

Yansheng Investment Management Co. LTD *(Stocks Quant Researcher)* *2023.11 - 2023.12*

- Mine Alphas based on stocks daily data and fundamental data, by reading research reports and papers.
- Build factors backtest framework using Python, with reference to factor_analyzer library, including factor processing, backtesting indicators calculation, hierarchical backtesting, etc.
- 11 alphas, 3 of excellent quality (5-day IC > 0.05, with great long returns and hierarchical returns).

CUMCM Mathematical Contest in Modeling *(Province Prize)* *2023.09*

- Analyze more than 900,000 sales data, complete data processing, achieve Clustering under day, month and year periods;
- Quantitate profit and cost using Polynomial Regression, predict future costs with ARIMA, establish profit maximization Nonlinear Programming Equation and solve it by Genetic Algorithm;
- Combined with the commodity correlation, the Integer-Nonlinear hybrid Programming Equation is constructed to select the commodities to achieve the maximum profit, which is solved by Monte Carlo method.

COMAP Mathematical Contest in Modeling *(M Prize)* *2023.02*

- Independently process 359 pieces of 12-dimensional data and explored additional 8 features;
- Establish ARIMA and BP NN to precisely predict the future daily activity of a game.
- Use MLR, Fisher Discrimination and Clustering to assess and classificate the words' difficulty, with 81.6% accuracy.

Skills and Honors

Programming:	Python | Pytorch | C | Git | MATLAB | R | LaTeX
Certifications	WorldQuant Alphathon Gold Certificate; WorldQuant Consultant; Coursera ML
Languages	IELTS 7 | CET - 6 - 612
Scholarships	National Inspiration Scholarship $\times$ 2